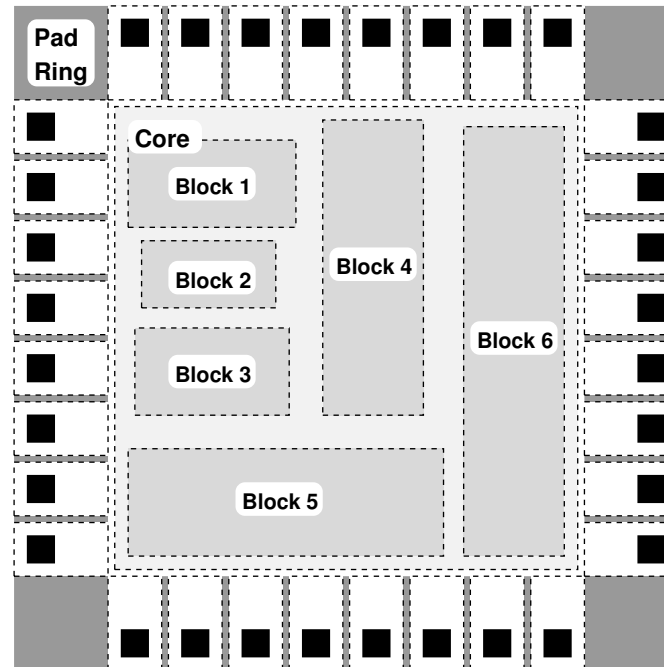
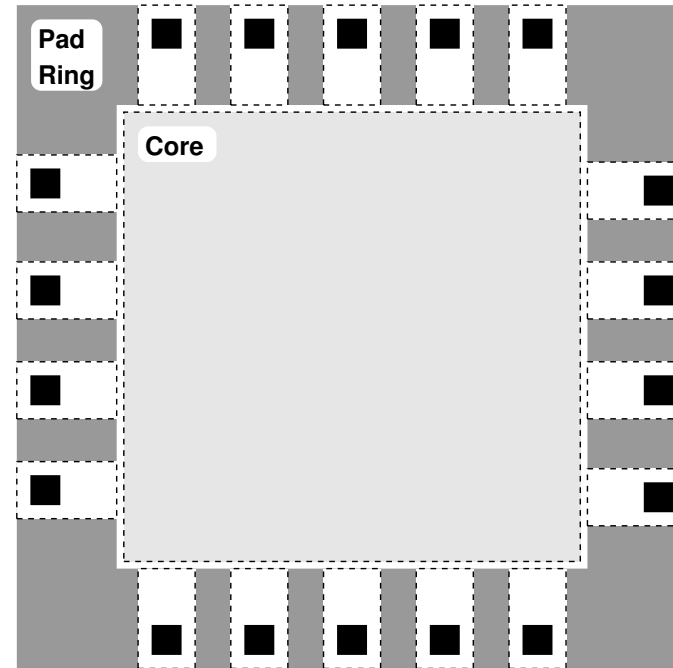
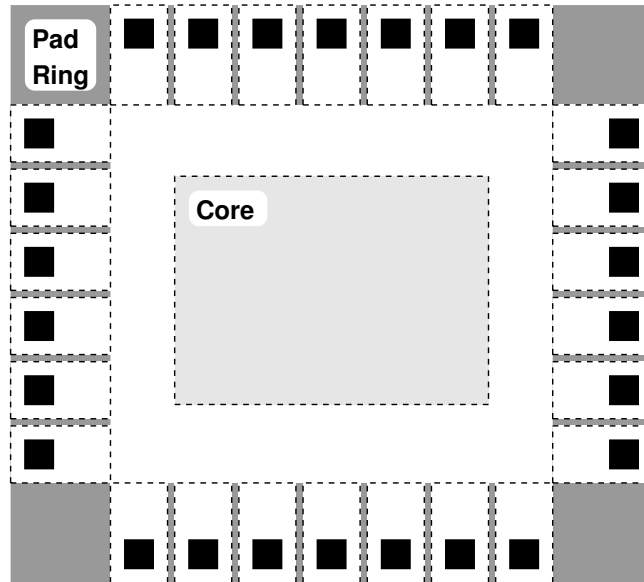


Pad Ring and Floor Planning



- The core of the chip (made up of one or more top level blocks) is surrounded by a ring of pads.
- The design of the blocks and the arrangement of blocks and pads can significantly affect the overall chip area (and hence the cost/yield).

Pad Ring



Pad Limited: small core and/or many pads

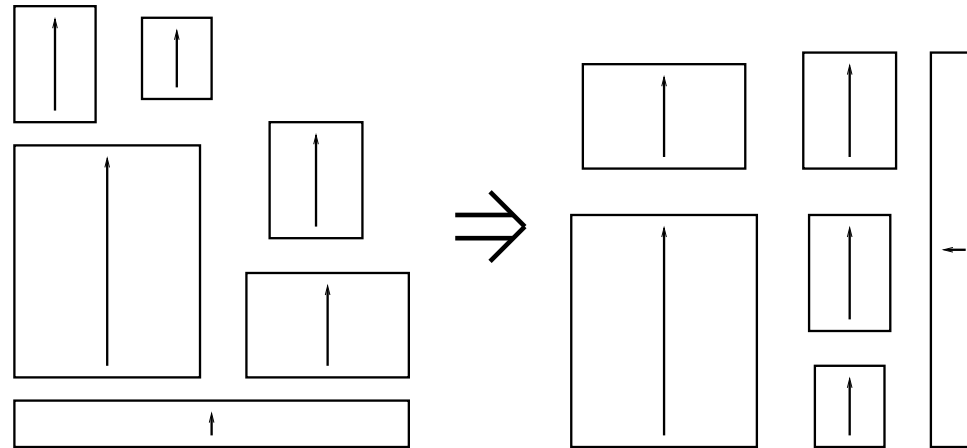
minimum pad to pad distance – gaps around core

Core Limited: large core and/or few pads

gaps between pads¹

¹these gaps will be filled with special filler cells

Floor Planning



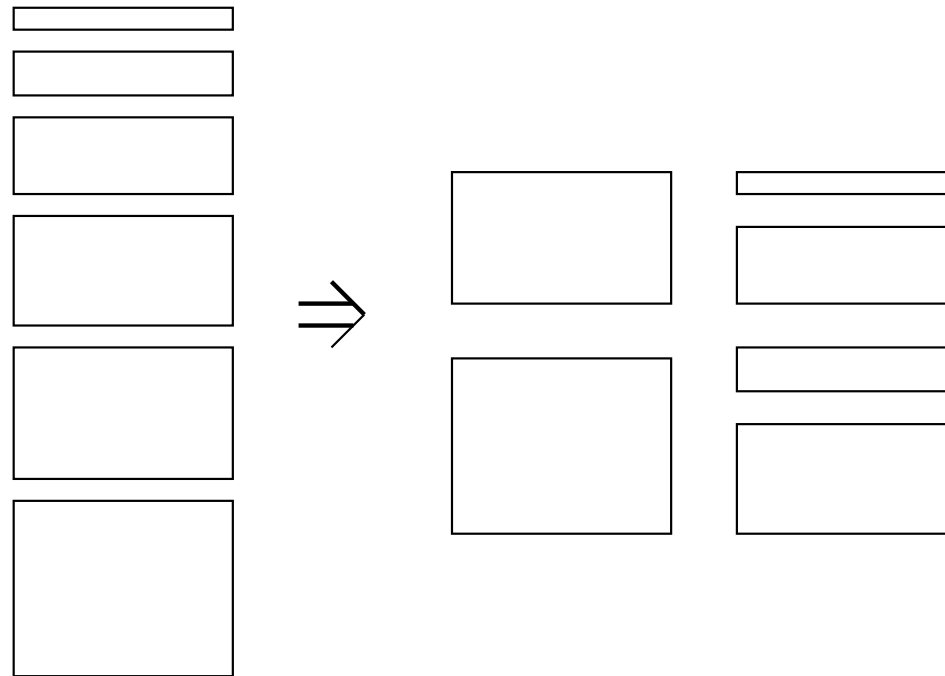
- Re-arrange and re-orient blocks to:
 - create a minimum number of major routing channels²
 - reduce block to block and block to pad routing

At top of the hierarchy, chips should be near square, other constraints exist at lower levels.

²for multi layer metal processes (≈ 5 metal layers or more) it should be possible to route over the blocks allowing closer placement

Block Design for easy Floor Planning

- Block shape



Where blocks share a common width, efficient placement is much easier.

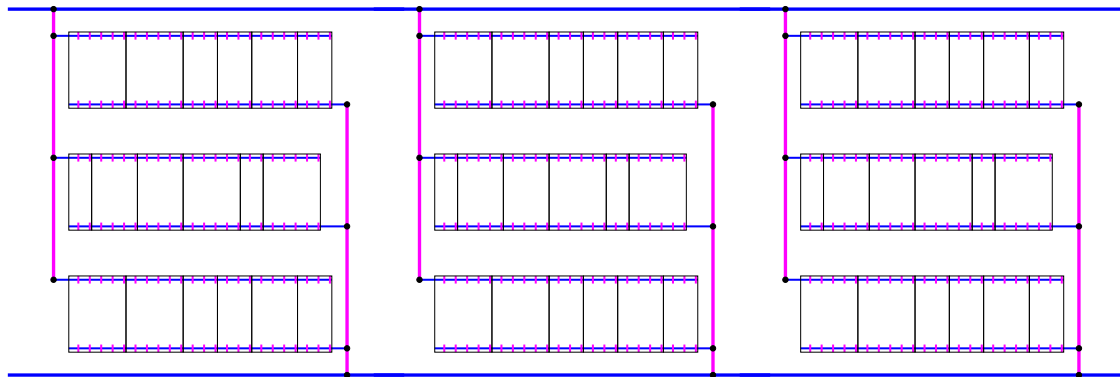
- Block ports

If possible arrange the ports on a block for ease of routing to pads and other blocks.

Floor Planning for Standard Cell Layout

Automatic layout:

- Flatten hierarchy.
- Placement is controlled by algorithms designed to minimize routing.
- Aspect ratio easy to control, also control number of columns and rows.



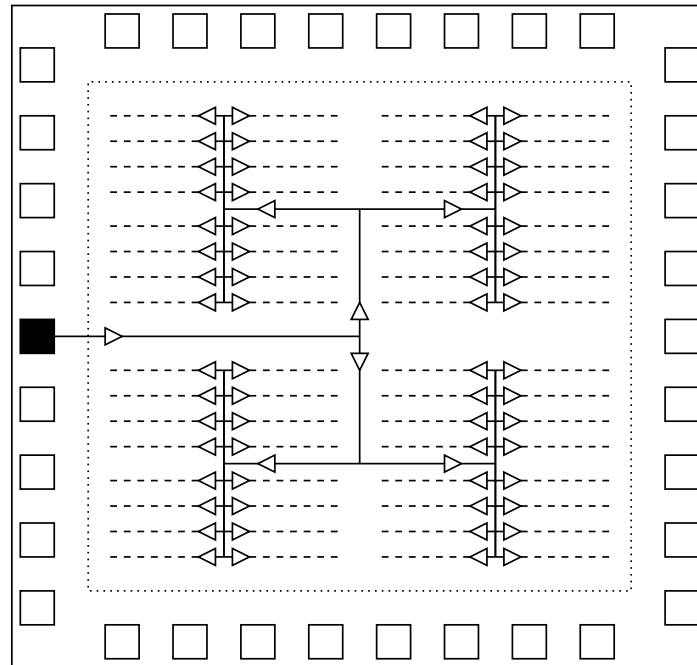
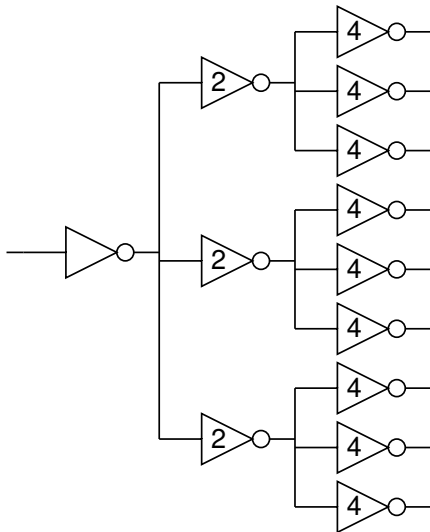
Manual layout:

- Placement based on layout hierarchy (essential for managing complexity).
- Aspect ratio and port position must be considered early as there is seldom time for iteration.

Global Routing

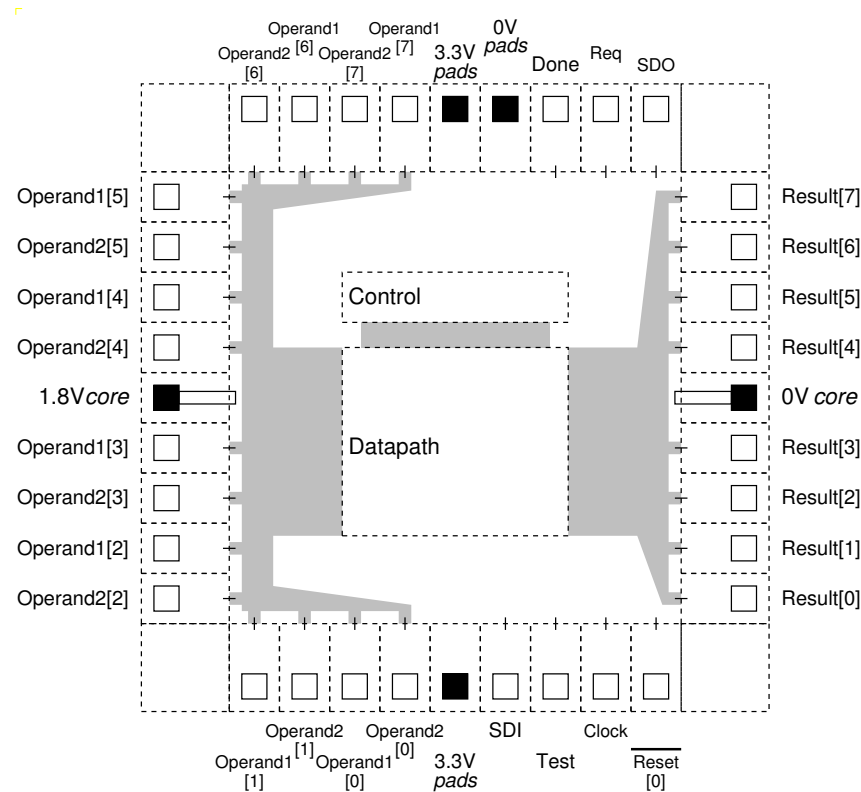
Route critical signals first.

- Buffer global and time critical signals.
- Clock distribution should be arranged to avoid skew across the chip³.



³buffering may actually increase delays while reducing skew

VLSI – Pad Ring and Floor Planning



- Pad ring pre-defined^a

create_pad_ring

-multiplier

<xcoreSize> <ycoreSize>

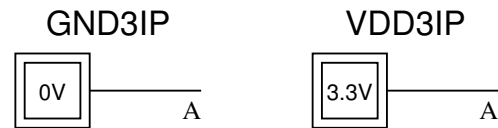
- Two blocks in core
 - Bitslice Datapath
 - Synthesized Control
- Pad limited
- Clock distribution built in to cell library

^adesign blocks to reduce routing
 since pads can't be moved

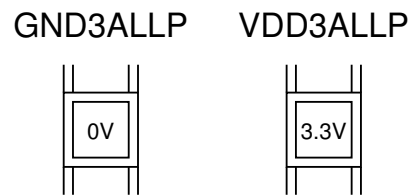
Datapath will be designed and placed to permit easy wiring of Operand and Result buses to left and right hand pads. Control will be designed and placed to permit easy wiring of control signals to the datapath.

VLSI – AMS 0.35 μ m CMOS Pads

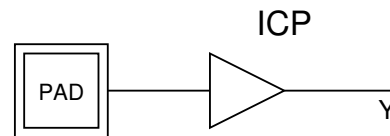
Core Power Supply Pads



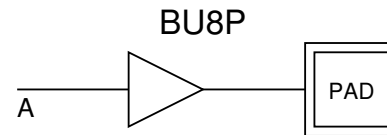
Pad Ring Power Supply Pads



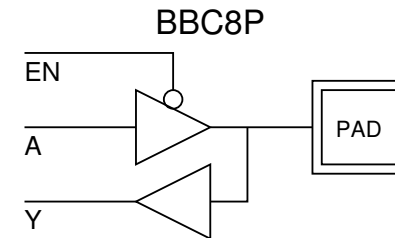
Input pad



Output pad



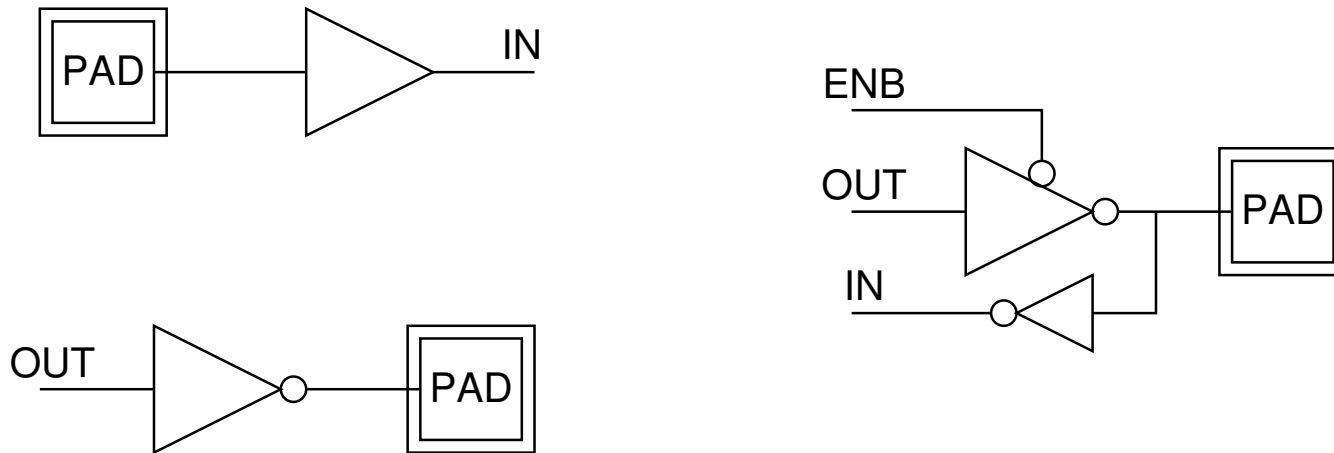
Bi-directional pad



- Large buffers on output pads allow for drive of very large external loads.
- Separate "dirty power" supply pads are provided for the main pad drive transistors to reduce switching noise in the core.
- Bi-directional pads require three connections to the core.

Input / Output

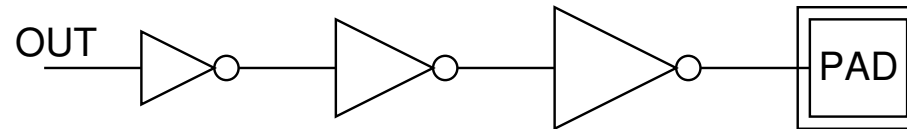
- I/O Pads



– A brief look at a selection of simple digital CMOS I/O pads

Output Pads

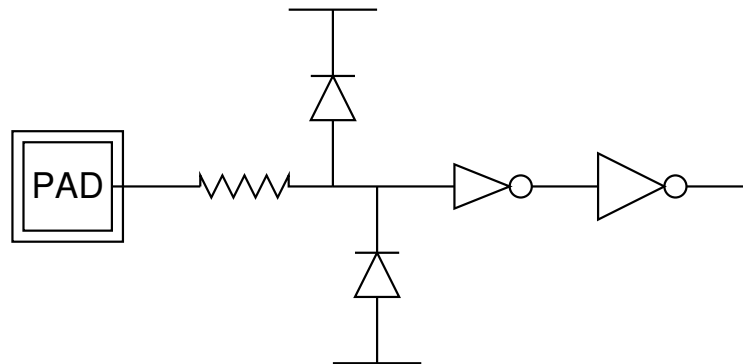
- Output pad driver



- ratioed inverters are used to provide appropriate drive capability
- final drive transistors are carefully designed to avoid latch-up
- pad rings are frequently powered separately (dirty power) to confine switching noise

Input Pads

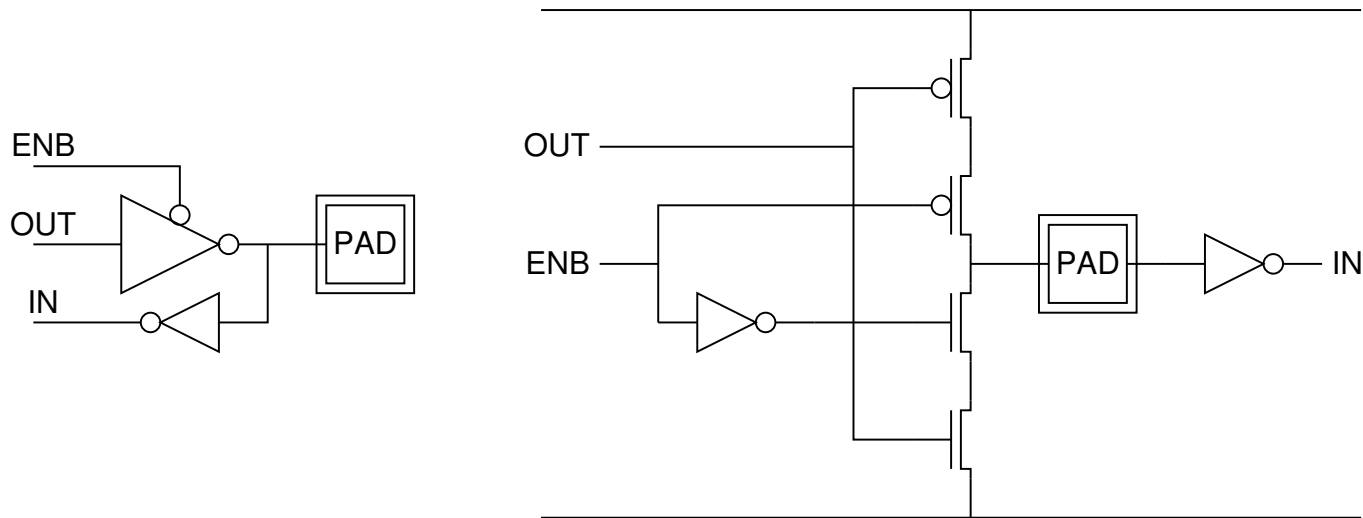
- Input protection



- must protect floating transistor gates from permanent damage via electrostatic discharge

Bidirectional Pads

- Simple bidirectional pad

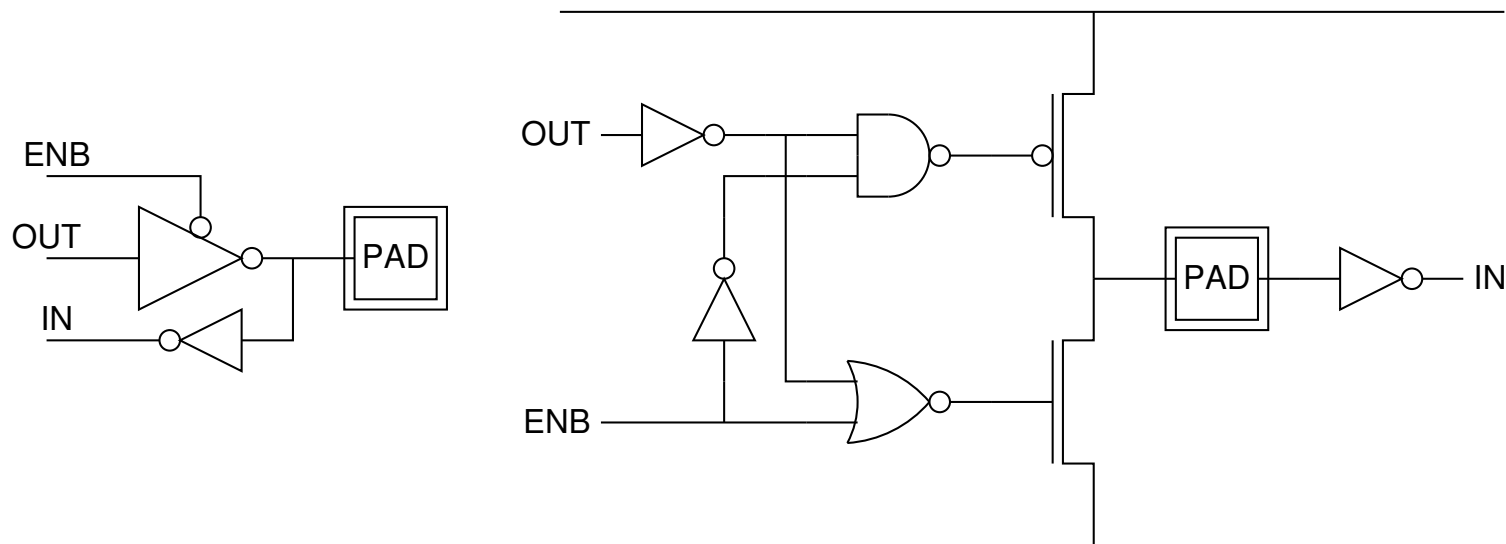


- bidirectional pad is a tristate inverter output driver combined with an input pad⁴
- even when IN and OUT are connected internally, we need buffering and an enable control signal

⁴note input protection is not shown here

Bidirectional Pads

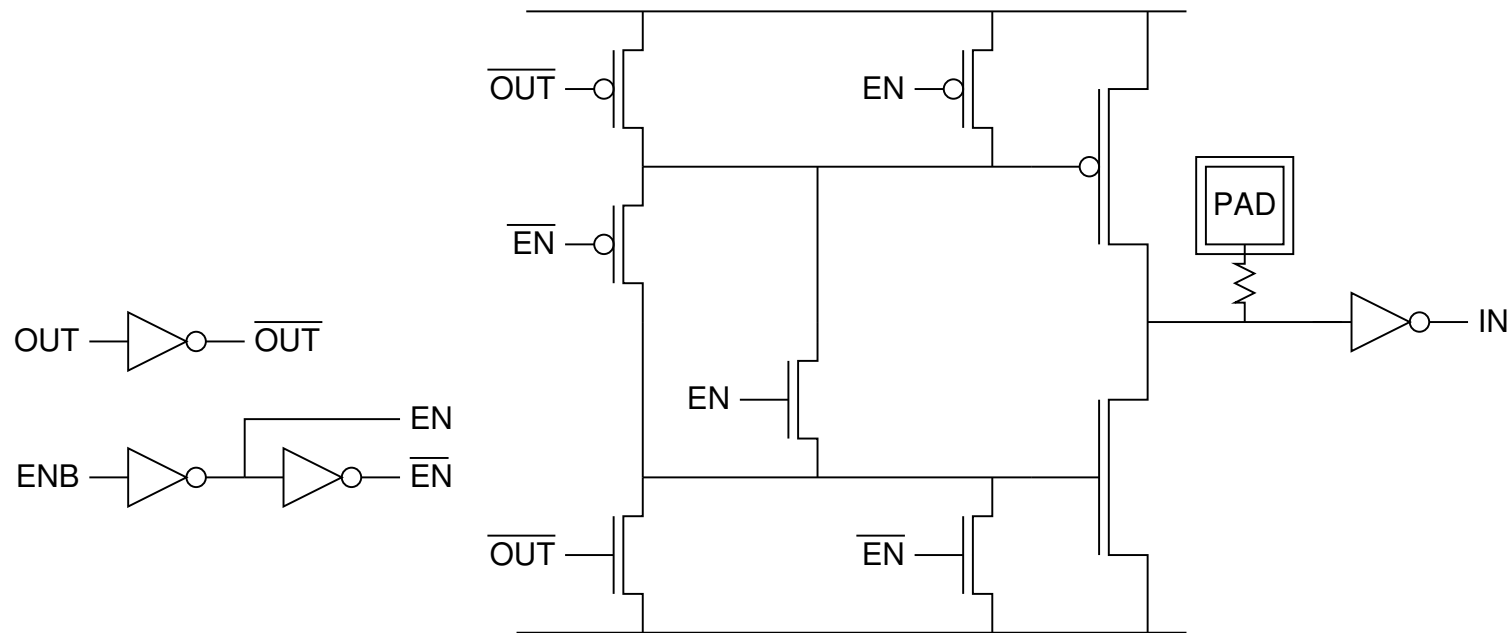
- Bidirectional pad with increased drive capability



– redesign to avoid series output transistors

Bidirectional Pads

- Advanced bidirectional pad design



- logic gates are merged
- output transistors act as diodes when not enabled
- low value diffusion resistor completes input protection circuit